Project Report Format

1. PROJECT TITLE

Strategic Retail Intelligence: A Business Analytics Dashboard for Product Positioning and Sales Performance Optimization

---

2. PROJECT DESCRIPTION (Detailed)

- **System Overview:** A centralized Business Intelligence (BI) platform built on Tableau that integrates multi-dimensional retail data. It moves beyond simple sales tracking to analyze *why* products perform differently based on their physical location in a store and their price relative to market competitors.
- **Problem Domain:** The project addresses the "Information Gap" in retail management, where managers have sales totals but lack visual evidence of how shelf-level factors (like being on an end-cap vs. a center aisle) affect different customer demographics.
- **Target Users:** Retail Category Managers, Marketing Strategists, and Inventory Planners.

---

3. APPLICATION SCENARIOS / USE CASES

- **Scenario A (Pricing Strategy):** A Pricing Manager uses the **Price Gap Analysis** scatter plot to identify "Overpriced Essentials." If a basic commodity is priced 15% higher than the competitor average and sales are dipping, the manager can use the dashboard to justify a price correction.
- **Scenario B (Marketing & Demographics):** A Marketing Lead notices via the **Demographic Donut Chart** that "Seniors" are the primary buyers of a specific premium coffee brand. They then check the **Positioning Heat Map** and realize that coffee is currently on a high shelf. They decide to move it to eye-level to further increase conversion for that specific age group.
- **Scenario C (Seasonal Inventory):** During the winter shift, the **Seasonality Filter** reveals that certain "Home Goods" move faster when placed near the entrance. The manager uses this insight to plan the floor layout for the upcoming holiday season.

---

4. TECHNICAL ARCHITECTURE OVERVIEW

Your architecture follows a **3-Tier Data Analytics Model**:

1. **Data Persistence Layer:** Consists of structured CSV/Excel flat files. This layer stores raw attributes: Product_ID, Shelf_Location, Current_Price, Competitor_Price, Units_Sold, and Customer_Segment.
2. **Business Logic Layer (Tableau Engine):** * **Data Transformation:** Cleans null values and standardizes category names.
    - **Calculated Fields:** Performs real-time math, such as [Price Gap] = [Our Price] - [Avg Competitor Price] and [Sales Efficiency] = [Units Sold] / [Foot Traffic].
3. **Presentation Layer (UI):** The final interactive dashboard hosted on Tableau Public/Desktop, featuring dynamic parameters and action filters.

---

7. PROJECT OBJECTIVES

- **Technical:** To develop a robust ETL (Extract, Transform, Load) pipeline that converts raw retail CSV data into optimized Tableau Hyper Extracts.
- **Analytical:** To provide a 360-degree view of sales performance by correlating at least four variables: Price, Position, Season, and Demographics.
- **Business:** To reduce the time spent on manual report generation by 70%, allowing for "live" strategic pivots during weekly performance reviews.

---

8. SYSTEM WORKFLOW (Step-by-Step)

1. **Data Acquisition:** Aggregating sales and competitor pricing into a standardized CSV format.
2. **Data Ingestion:** Connecting Tableau to the data source and defining metadata (Dimensions vs. Measures).
3. **Logical Processing:** Creating sets and groups for "High-End" vs. "Budget" products and calculating the "Price Index."
4. **Visual Construction:** Building individual sheets (Heat Maps for aisles, Trend lines for seasonality).
5. **Dashboard Integration:** Using "Dashboard Actions" so that clicking a segment in the Donut Chart automatically filters the Heat Map.

6. **Insights Delivery:** Reviewing the finalized dashboard to extract actionable business recommendations.

---

11. IMPLEMENTATION & FEATURES

- **11.1 Competitive Benchmarking (Scatter Plot):** Visualizes the "Price Gap." Products in the top-left quadrant are identified as "High Risk" (High Price / Low Sales), while the bottom-right identifies "Market Leaders."
- **11.2 Shelf Positioning Analysis (Heat Map):** Uses a color gradient (Red to Green) to show which aisle locations are "Hot Zones" for revenue generation.
- **11.3 Demographic Deep-Dive (Donut Charts):** Breaks down the "Volume of Sales" by age group and gender, allowing for hyper-localized marketing strategies.

---

17. ADVANTAGES & LIMITATIONS

- **Advantages:** * **Evidence-Based:** Replaces "manager's intuition" with hard data.
    - **High Interactivity:** Users can "slice and dice" data without knowing SQL or complex coding.
- **Limitations:**
    - **Static Nature:** Currently relies on manual CSV uploads rather than a live API feed.
    - **Historical Data:** It describes what *has* happened; it does not yet include a Machine Learning "Predictive" model for future sales.

---

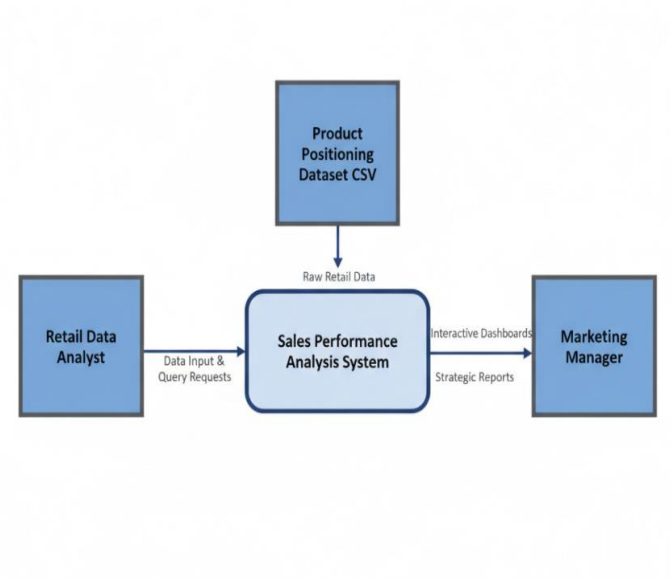
19. CONCLUSION

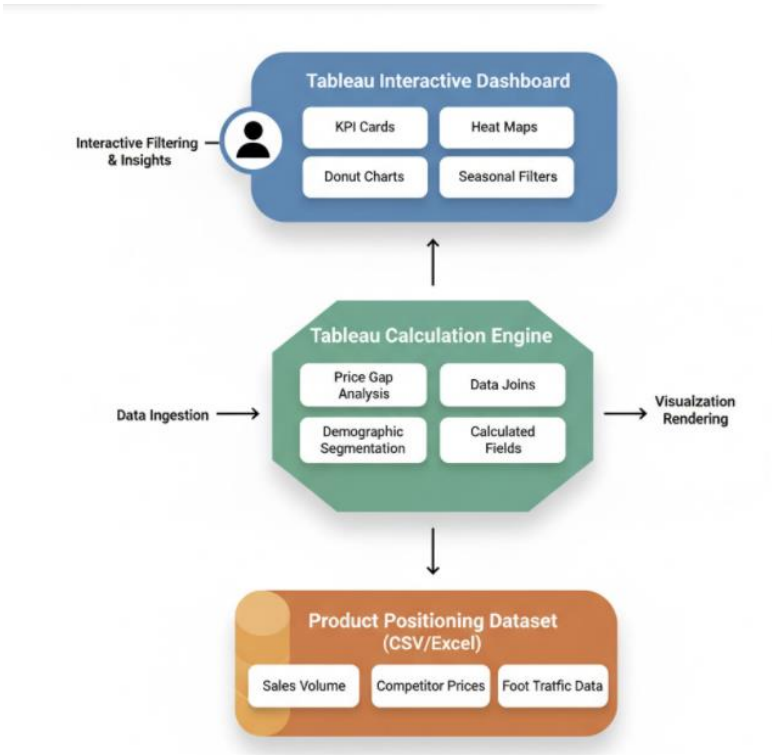
The **Product Positioning & Sales Performance Analysis** project successfully demonstrates the power of visual analytics in the retail sector. By bridging the gap between raw datasets and executive-level decision-making, the system provides a clear roadmap for optimizing store layouts and pricing structures to maximize profitability.

---

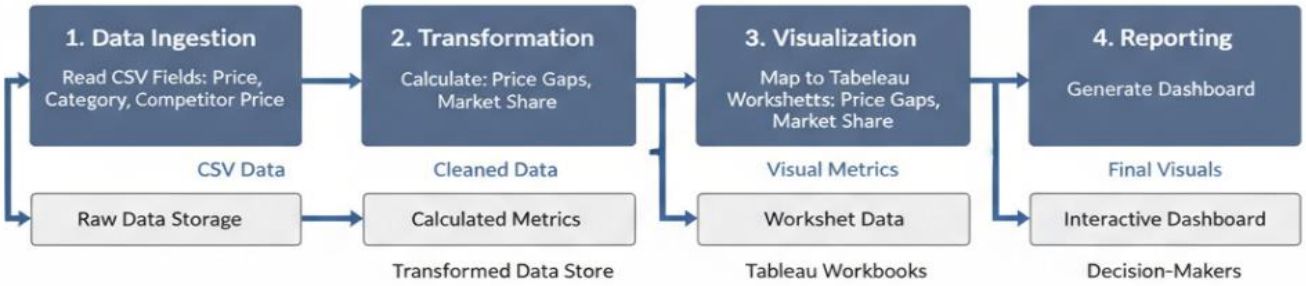
20. APPENDIX

- **20.1 Source Code & Project Files:**
    - **GitHub Repository Link:** [Insert your GitHub URL here]
    - **Contents:** The repository includes .twb or .twbx (Tableau Workbook) files, the cleaned dataset, and all project documentation PDFs.
- **20.2 Dataset Details:**
    - **Source:** The data is a "Retail Product Positioning & Sales Dataset."
    - **Format:** CSV (Comma Separated Values).
    - **Key Fields:** The most important columns used for logic: Product_ID, Category, Store_Location, Shelf_Position, Price, Competitor_Price, Units_Sold, and Customer_Demographics.
- **20.3 Design Assets:**
    - The **Data Flow Diagrams (DFD)** and **Technical Architecture** diagrams were designed using **Miro**.





-


1. Data Ingestion
Read CSV Fields: Price, Category, Competitor Price
2. Transformation
Calculate: Price Gaps, Market Share
3. Visualization
Map to Tabeleau Worksheets: Price Gaps, Market Share
4. Reporting
Generate Dashboard
CSV Data
Cleaned Data
Visual Metrics
Final Visuals
Raw Data Storage
Calculated Metrics
Workshet Data
Interactive Dashboard
Transformed Data Store
Tableau Workbooks
Decision-Makers